SERVICE REQUEST AND DELIVERY:
EVIDENCE FROM 311 SYSTEM IN CHICAGO

PPHA 30538 Data Analytics and Visualization for Public Policy
Final Project

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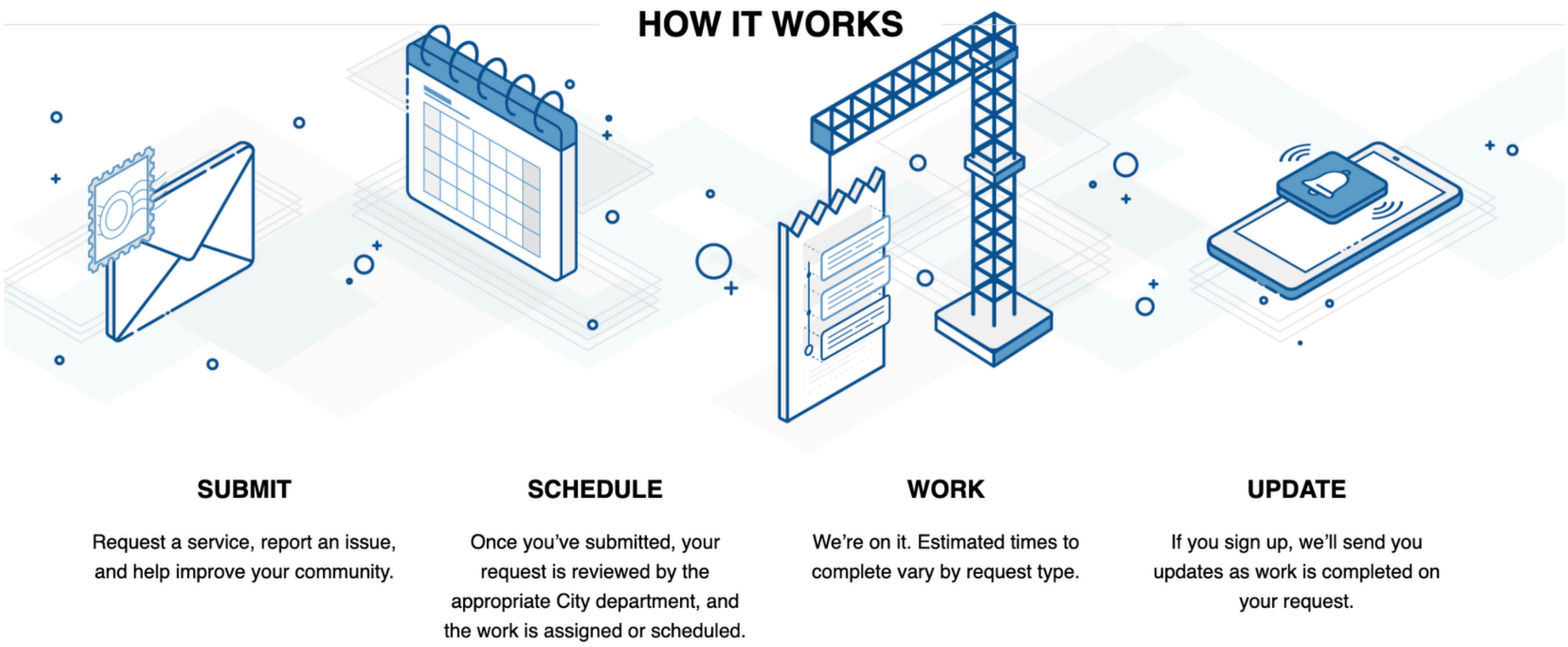
CONTEXT:

Chicago exhibits substantial variation in median household income across neighborhoods.

Public service delivery, such as street maintenance and pothole repairs, may also vary across space, raising concerns about whether municipal services are distributed equitably.

The **City of Chicago's 311 system** provides a centralized platform through which residents submit service requests. These data offer a unique opportunity to evaluate how the city responds to neighborhood needs.



MOTIVATION:

In front of **International House on East 59th Street**, there is a visible pothole accompanied by an **active 311 service request**. Despite being located near the University of Chicago (an area often perceived as relatively advantaged) the issue is **unresolved.**

This observation raised an important question:

If a service request in a relatively higher-income area experiences delays, what happens in lower-income neighborhoods?

Service Request # :

Pothole in Str

1411 E 59TH ST

Chicago Illinois 60637

QUESTIONS & OBJECTIVE:

RESEARCH QUESTION

Are neighborhood income differences in Chicago associated with disparities in 311 service demand and response times across Community Areas?

More specifically:

- Is service demand systematically different across income groups?
- Do higher-income neighborhoods experience faster completion times?
- Are there patterns suggesting inequitable municipal responsiveness?

OBJECTIVE

To examine whether neighborhood income levels in Chicago are associated with differences in 311 service demand and response times across Community Areas.

DATA SOURCES:

311 Service Requests
Source: City of Chicago Data Portal (CHI311)

- Universe of non-emergency municipal service requests
- Includes request type, location, submission date, and completion date
- Enables measurement of:
  - Service demand (volume per Community Area)
  - Service efficiency (response and completion time)

American Community Survey (ACS) 5-Year Est.
Source: U.S. Census Bureau – ACS 5-Year Data (Community Area level)

- Provides neighborhood-level socioeconomic indicators
- Median household income used to group neighborhoods
- Captures structural income differences across Chicago
- Allows classification of Community Areas into income quantiles

DATA PROCESING:

Sample construction

Filtered 311 service requests submitted between January 1, **2023** – December 31, **2024**

Key Variables: Creation date, Completion date, Service request type, Status, Community Area

Outcome Variable

Response Time (days) = Completion Date − Creation Date

Aggregation

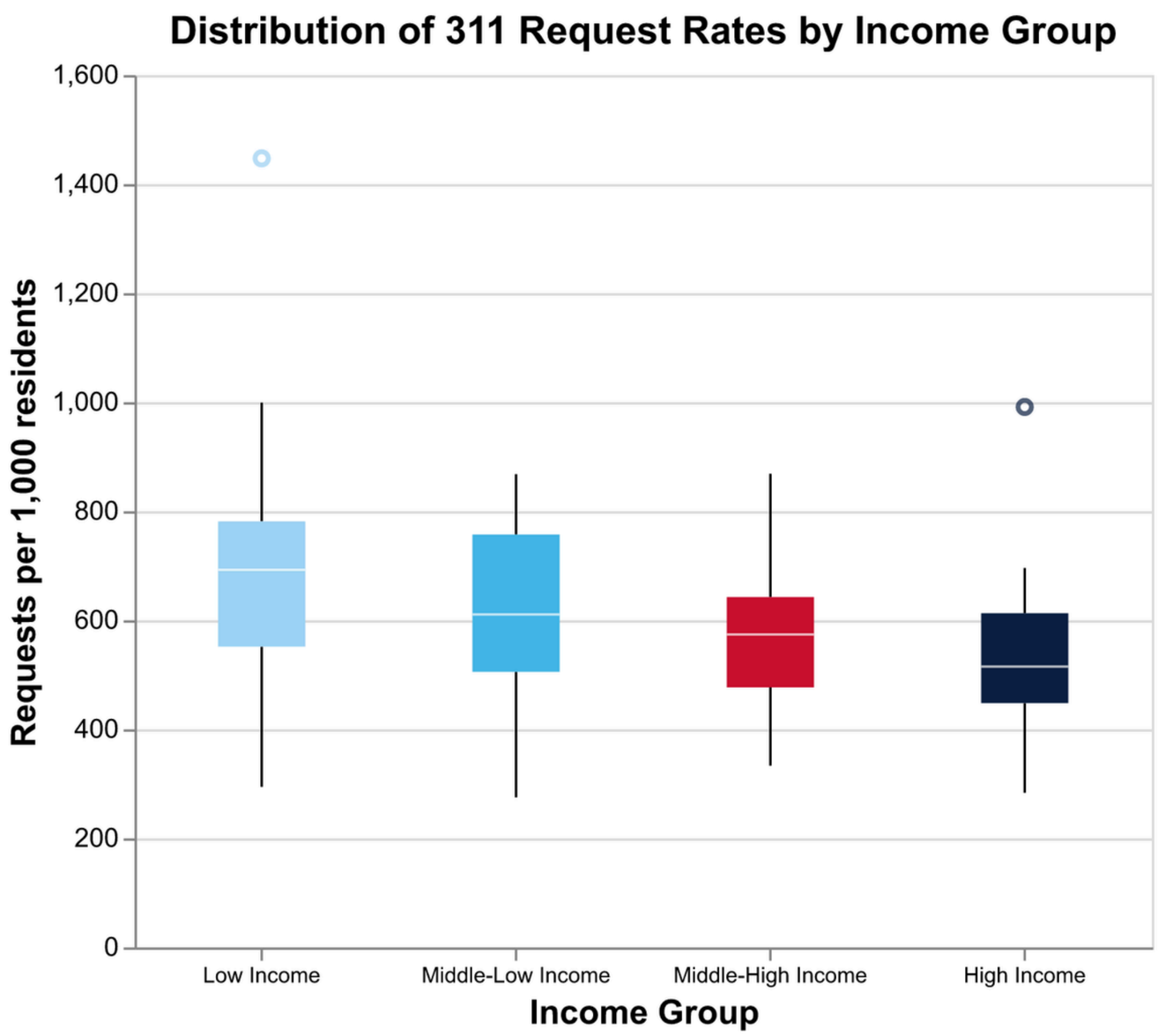
Computed average response time and total request volume at the Community Area level

Socioeconomic Integration

Merged with 2019–2023 ACS 5-Year Median Household Income

Grouped Community Areas into Income Quartiles (Q1 Low to Q4 High income)

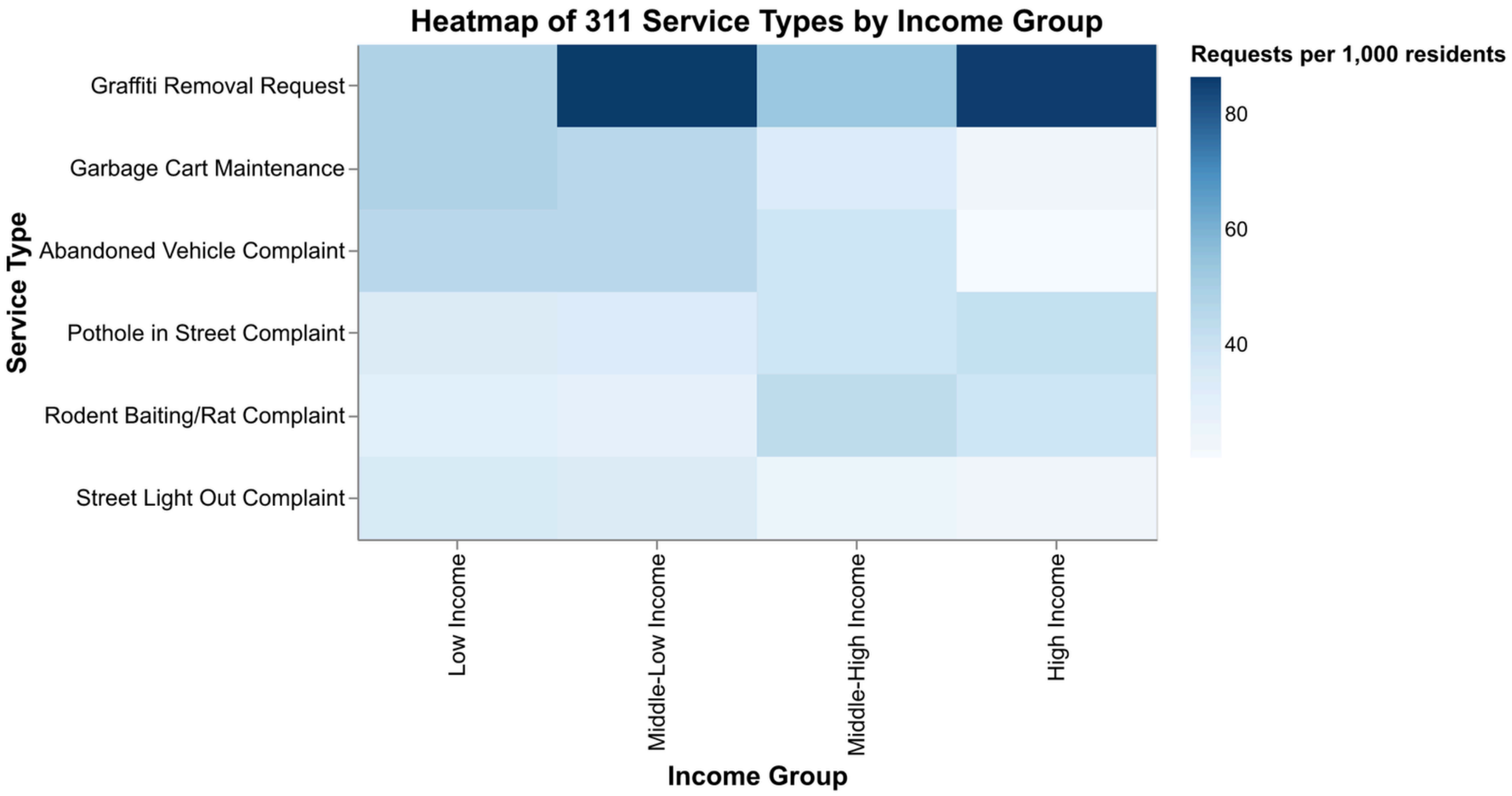
KEY FINDINGS #1:



311 request are higher in lower-income communities, greater concentration of reported service needs.

- Median request rates decline as neighborhood income increases.
- Variation within income groups suggests heterogeneity across neighborhoods

KEY FINDINGS #2:

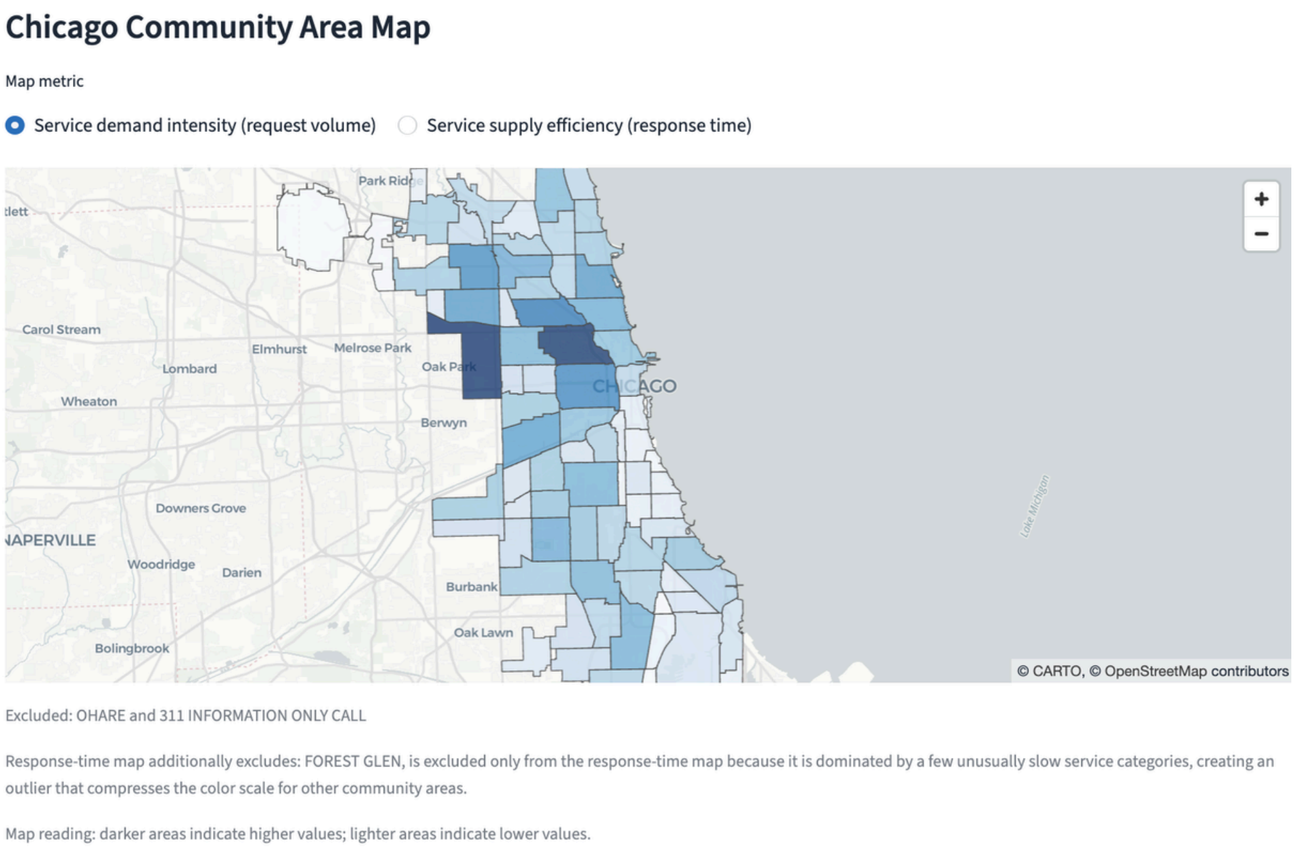


311 service request patterns vary by neighborhood income level

- Abandoned vehicle reports are more prevalent in **lower-income** areas.
- Graffiti removal requests are more common in **higher-income** neighborhoods

KEY FINDINGS #3:

CTRL + CLICK ON THE IMAGE



POLICY IMPLICATIONS:

- The City can use 311 data to identify delays in service response processes and **redesign workflows** to reduce average completion times across high-demand service categories.

- Historical 311 request patterns can inform **predictive maintenance programs**, allowing the City to anticipate recurring issues, such as potholes or rodent complaints, and address them before formal service requests accumulate.

- Developing internal real-time monitoring dashboards based on 311 data could help managers track workload, response times, and service backlogs, enabling **faster operational adjustments**.

- Incorporating 311 service metrics into budgeting decisions can help align **municipal spending with measurable performance indicators**, improving accountability and operational effectiveness.

CHI311

QUESTIONS?

RECOMMENDATIONS?

Thanks!